

One step at a time

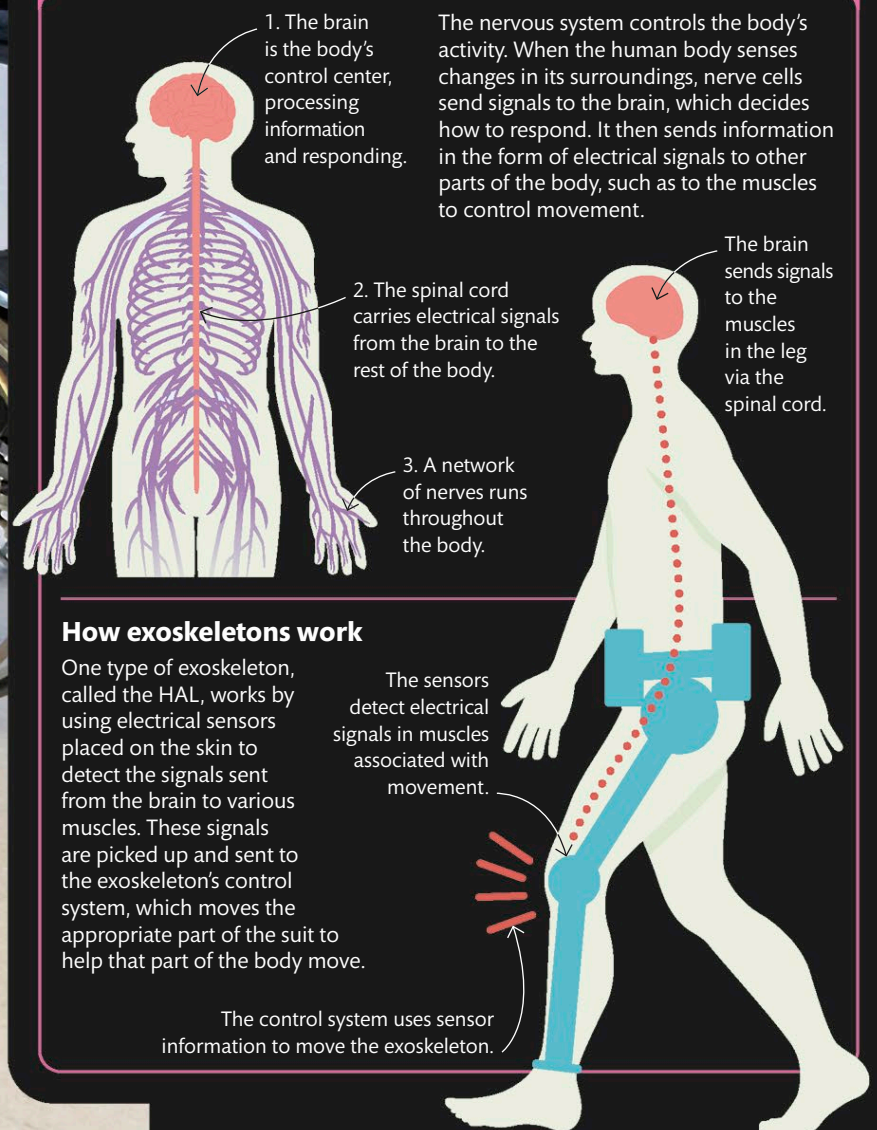
At the Cyberdyne facility in Nordrhein-Westfalen, Germany, people with damage to their lower nervous systems are using the HAL exoskeleton to help them regain mobility. The exoskeleton supports the movements of the patient's limbs and then feeds back to the brain, strengthening the connection between the brain and the limb.

MOBILITY MACHINE

EXOSKELETONS

In nature, exoskeletons are armorlike structures that support the bodies of animals such as insects. Artificial human-made exoskeletons work similarly—they are wearable machines that support and enhance how the body moves using a system of sensors, motors, levers, and other parts. Some exoskeletons can sense the signals sent by the body's nervous system and interpret these, while others work using sensors in the shoes that activate when a person's body weight shifts.

THE NERVOUS SYSTEM



In 2020, American Adam Gorlitsky ran a 33-hour marathon while wearing a type of exoskeleton.